

Radar HRRP Automatic Target Recognition: Algorithms and Applications

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Abstract

Radar automatic target recognition (RATR) is an important function for modern radar. Target high resolution range profile (HRRP) contains target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the RATR community. Statistical modeling of target HRRPs is the key stage for HRRP statistical recognition, which including model selection and parameter estimation. In this paper, we introduce the recent progress in HRRP based RATR, including the statistical modelling algorithms of HRRP data, robust recognition algorithms for low signal noise ratio(SNR) case, and some results based on the measured data.

Keywords: radar target recognition, high resolution range profile, factor analysis, local factor analysis, Temporal Factor Analysis

1. Introduction

A target high-resolution range profile (HRRP) is the amplitude of coherent summations of the complex time return from target scatterers in each range cell, as shown in Fig.1, which is shown to be promising signature for RATR [1-7]. Statistical recognition methods have been extensively studied and successfully applied to HRRP-based RATR area [1-7]. By statistical recognition we mean the class membership of a measured HRRP is determined by its posterior probabilities of each class. For HRRP-based statistical recognition, one key problem is to choose an appropriate model that can describe HRRP's statistical property accurately. There are several issues need to be considered. The first is that the data between range cells are statistical correlated or not. The second is that the data follow which kind of distribution model. The third is that how to build statistical model for the HRRP data according to different target orientations. In RATR applications, the amplitude of received HRRP signal is a function of radar-target distance, namely, the SNR of HRRP signal will be much lower for the target at longer distance. Therefore, it is very important for the RATR algorithms have a robust performance for the low SNR data. In other words, to obtain satisfactory recognition performance for longer distance.

The remaining of this paper is organized as follows. In section 2, we give some background knowledge about HRRP-based target recognition. Section 3 introduces the statistical models for HRRP data. In section 4, we introduce the robust recognition algorithms for low SNR data. Based on measured

HRRP data, some results are presented in section 5. Finally, some remarks and discussions are drawn in section 6.

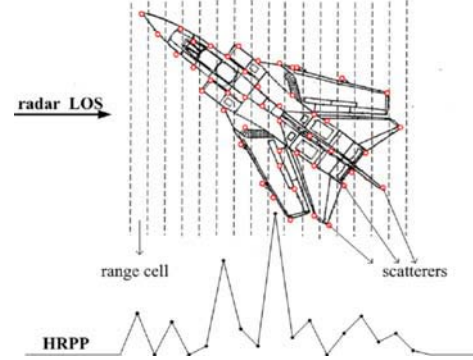


Fig. 1. Radar returns from the scatterers on the target are projected onto the LOS, resulting in an HRRP.

2. Statistical modelling of radar HRRP data

The statistical modelling of radar HRRP data includes the model selection and parameters estimation. In this section, we will introduce the recent progress of the statistical modelling algorithms for HRRP data.

2.1. Statistical independent Gaussian Model

Based on the assumptions that the HRRP data are Gaussian distributed, and statistical independent between different range cells. Therefore, we can estimate the mean vector and the covariance matrix (actually is the covariance vector under this assumption.) as

$$\hat{\mathbf{m}}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} \mathbf{x}_{j,i} \quad (1)$$

$$\hat{\Sigma}_i = \text{DIAG} \left[\frac{1}{N_i} \sum_{j=1}^{N_i} (\mathbf{x}_{j,i} - \mathbf{m}_i)^T (\mathbf{x}_{j,i} - \mathbf{m}_i) \right] \quad (2)$$

Where $\mathbf{x}_{j,i}$ and N_i are the j th HRRP data of i th target and the training data number, respectively. Based on the estimated parameters, one can make a decision for a test data sample by

$$D_i = \det[\hat{\Sigma}_i]^{-1/2} + (\mathbf{x} - \mathbf{m}_i)^T \hat{\Sigma}_i^{-1} (\mathbf{x} - \mathbf{m}_i) \quad (3)$$

2.2 Statistical independent Non-Gaussian Model

For wideband radar, it is well known that the target echoed signal is generally Non-Gaussian distributed. Therefore, the Gaussian distributed assumption can not get a satisfactory recognition performance. In [1] and [2], Gamma distribution is used to describe the HRRP data. Actually, the distribution model of HRRP data in different range cells maybe different. We divide the range cells into three types according to different scatterer distribution structures [3]:

a) First type of range cell: There are a large number of small scatterers and no predominant scatterers in this type of

range cell. Under the assumption that the intensities of the small scatterers are almost same and the number of the small scatterers is large enough, the central limit theorem holds. The inphase or quadrature component of the complex echo of this type of cell will follow Gaussian distribution with zero mean and its amplitude will follow Rayleigh distribution as

$$P_{\text{Rayleigh}}(x) = \begin{cases} \frac{x}{b^2} \exp\left(-\frac{x^2}{2b^2}\right) & x > 0 \\ 0 & x \leq 0 \end{cases} \quad (4)$$

where b is the scale parameter;

b) Second type of range cell: This type of range cell consists of a predominant scatterer and a large number of small scatterers. Under the hypothesis similar to the first type, the inphase or quadrature component of the complex echo of this type of cell will follow Gaussian distribution with the mean determined by the echo's inphase or quadrature component of the predominant scatterer, in accordance with the central limit theorem. Thus, its amplitude follows a Ricean distribution amplitude as

$$P_{\text{Rice}}(x) = \begin{cases} \frac{x}{b^2} \exp\left(-\frac{x^2 + |v|^2}{2b^2}\right) I_0\left(\frac{x \cdot |v|}{b^2}\right), & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (5)$$

where b is the scale parameter, v is the noncentrality parameter, and $I_0(\bullet)$ is the modified Bessel function of zero order.

c) Third type of range cell: There are a large number of small scatterers and several predominant scatterers in this type of range cell. its statistical property can be approximately represented by a multimodal distribution, which depends on the particular geometrical distribution of the predominant scatterers.

We can use Gamma distribution to describe the statistical property of the first type and second type of range cell, and Gaussian Mixture model to describe the third type of range cells.

The next question is that how to determine the range cells type by give the training dataset. We use the rival penalized competitive learning (RPCL) algorithm [8], which is an efficient clustering algorithm, to determine the type of a range cell. And the parameters of Gamma distribution and Gaussian mixture distribution are estimated by the maximum likelihood (ML) method and the expectation-maximization (EM) algorithm, respectively [3].

2.3. Statistical correlated Gaussian Model

Theoretically speaking, the target can be regarded as consist of multiple isolated scatterers, therefore, the HRRP data can be assumed as statistical independent between range cells. However, considering the multiple reflections phenomena, target structure similarity between range cells, etc, this independence assumption is not true. One can use joint Gaussian distribution directly to model the range cells correlation property of HRRP data. However, the model parameter dimension will be increased significantly. This will introduce difficulty in parameter estimation, larger memory and computational complexity requirement. Therefore, lower dimension model is required.

a) Probability Principal Components Analysis Model (PPCA) Model): For a HRRP sample x with dimension of d , it can be written as:

$$x = A_k y_k + \mu_k + \varepsilon_k \quad (6)$$

Where k is target label, μ_k is a d -dimensional mean vector, $A_k \in \mathbb{R}^{d \times q}$ is the weight matrix with column orthogonal to each other, the latent variable $y_k \in \mathbb{R}^q$ follow the Gaussian distribution and $\varepsilon_k \in \mathbb{R}^d \sim N(0, \sigma_k^2 I_d)$ $\varepsilon_k \in \mathbb{R}^d$ is the Gaussian distributed noise variable. Therefore, the distribution of x can be written as:

$$p(x|k) = (2\pi)^{-d/2} |\sigma_k^2 I_d + A_k A_k^T|^{-1/2} \exp\left\{-\frac{1}{2}(x-\mu_k)^T (\sigma_k^2 I_d + A_k A_k^T)^{-1} (x-\mu_k)\right\} \quad (7)$$

b) Factor analysis (FA) Model: FA model has the same formation with PPCA model as shown in (6). PPCA model requires orthogonal bases in the signal subspace and independent elements in the noise variable with same variances, while FA model requires independent bases in the signal subspace and independent elements in the noise variable with different variances. In FA model, the noise variable $\varepsilon_k \in \mathbb{R}^d \sim N(0, \psi_k)$, ψ_k is a diagonal matrix with different elements. Therefore, the distribution of x can be written as:

$$p(x|k) = (2\pi)^{-d/2} |\psi_k + A_k A_k^T|^{-1/2} \exp\left\{-\frac{1}{2}(x-\mu_k)^T (\psi_k + A_k A_k^T)^{-1} (x-\mu_k)\right\} \quad (8)$$

Our analysis results show that the recognition performance of FA model is better than that of PPCA model [5]. Detail approaches about the model selection and parameters estimation methods can be found in [5].

2.4. Statistical correlated Non-Gaussian Model

As indicated before, the HRRP data is generally non-Gaussian distributed, FA model actually is a approximately model to describe the statistical property of HRRP data. Consider that the statistical correlated between range cells, a statistical model which can describe the statistical correlated non-Gaussian HRRP data is desired.

We introduce local factor analysis (LFA) model into radar HRRP recognition. In LFA model, the data can be represented as the summation of multiple FA model:

$$p(x|k) = \sum_{l=1}^L \alpha_l p(x|k, l) \quad (9)$$

Where $\forall \alpha_l \geq 0, \sum_{l=1}^L \alpha_l = 1$. In FA model, the latent variable dimension need to be determined by model selection algorithm, and the loading matrix A_k , the statistical parameters of $\mu_k, y_k, \varepsilon_k$ need to be estimated. For LFA model, the model selection and parameters estimation is much more complex than that of FA model. Due to space limit, the detail approach is not introduced here, one can find them in [6].

2.5 Statistical Model by using temporal correlation

In applications, radar generally can obtain multiple HRRP during a period of time. One can use multiple HRRP observations together to improve the recognition performance. One direct approach is to regard the multiple HRRP data as statistical independent, and fusion the recognition results of each HRRP data together to get the final result. However,

considering that target-radar orientation varies continuously during the observation, the HRRP sequence may be statistical correlated. It is expected that the recognition performance can be further improved by using the correlation property between HRRP data. Motivated by which, we introduce the Temporal Factor Analysis (TFA) model [10,11] for HRRP recognition application. In this model, the HRRP data can be written as

$$\begin{aligned} \mathbf{x}_{t,k} &= \mathbf{A}_k \mathbf{y}_{t,k} + \boldsymbol{\mu}_k + \varepsilon_{t,k}, \quad \mathbf{y}_{t,k} = \tilde{\mathbf{B}}_k \mathbf{y}_{t-1,k} + \boldsymbol{\omega}_{t,k}, \quad t=1,2,\dots,T \\ \varepsilon_{t,k} &\sim N(\varepsilon_{t,k} | \boldsymbol{\theta}, \boldsymbol{\Psi}_k), \quad \boldsymbol{\omega}_{t,k} \sim N(\boldsymbol{\omega}_{t,k} | \boldsymbol{\theta}, \boldsymbol{\Omega}_k) \end{aligned} \quad (10)$$

where time is indexed by discrete t , parameters $\mathbf{A}, \boldsymbol{\mu}, \boldsymbol{\Psi}$ have the same meaning as in (6), ε_t and $\boldsymbol{\omega}_t$ are assumed to be uncorrelated random noises, ε_t is assumed to be uncorrelated, and $\mathbf{y}_t$ and $\boldsymbol{\omega}_t$ are uncorrelated with $\mathbf{y}_{t-1}$. Moreover, $\tilde{\mathbf{B}}$ is an $m \times m$ transition matrix, The temporal relation is embodied in the first-order vector autoregressive process in the hidden state sequence $\{\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_T\}$. It should be noted that the model parameters $\mathbf{A}, \boldsymbol{\mu}, \tilde{\mathbf{B}}, \boldsymbol{\Psi}, \boldsymbol{\Omega}$ are time invariant. Based on the further constraints that $\tilde{\mathbf{B}}, \mathbf{A}, \boldsymbol{\Psi}$ are all diagonal matrices, in [7] a model selection and parameters estimation approach is proposed.

3. Robust recognition in low SNR situation

Almost all the existing statistical models for HRRP data assume that the training data and test data are obtained under the similar measurement circumstance. In other words, the test data and training data share the same distribution model and same statistical parameters. In real application, the training data are usually collected under the condition of SNR via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance, where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large distance between the non-cooperative targets and radar. To handle this problem, one method is to improve the SNR at signal processing stage. However, for radar application, it is always the first desirability to find the target and recognition it at longer distance. Therefore, the robust recognition algorithm at low SNR situation is desired.

For the simple statistical models used in [1,2,3], since there is no model variable to describe the noise term, it is difficult to represent different noise background. For the FA model, from equation (6) we can see that the model parameters $\mathbf{A}_k, \boldsymbol{\mu}_k$ is determined by the signal of interest component, and the statistical parameters of latent variable $\mathbf{y}_k$ is also determined by the signal component. Only the statistical parameters of ε_k is determined by the noise. In other words, for different SNR level, if the statistical parameters of ε_k are updated, the model can be used for different SNR cases. The conclusion also applies to the LFA and TFA models.

Motivated by the above analysis, we developed a noise adaptive HRRP statistical recognition algorithm. The statistical model is trained by the high SNR data. When a test data come, we first keep the signal related parameters

unchanged, use the test sample to estimate the noise related statistical variable. Then using the updated statistical model with the estimated noise variable to perform the recognition.

4. Application issues

In application stage, there are several issues need to be considered:

a) Target-Aspect Sensitivity

According to the target scattering center model, for high resolution radar system, an aircraft-like target does not appear as a "point target" any longer, but consists of many scatterers separated in several range cells along the radar LOS. The scatterer distribution of a target will vary with the target-aspect, because the variation of target-aspect causes some scatterers to remove from one range cell to another. Namely, the HRRPs according to different target aspects are different. To handle this problem, we can divide a target into several aspect frames, in each frame, it is assumed that the HRRP follow same statistical model. Therefore, we can train multiple models for each target corresponding to different target aspect frames. Naturally, another question may be raised. How to divide the aspect frame? Actually, jointly with the statistical model training, the target aspect frame partition can be accomplished at same time. One can go to [3-6] for detail.

b) Time-Shift Sensitivity

Since we can not measure the target's distance accurately, the HRRP data in the sampling window may exist time shift. The time shift should be compensation during training and test stage.

c) Amplitude-Scale Sensitivity

If training samples and test samples are measured under same conditions, the amplitude-scales of HRRPs, which contain the target radar cross section (RCS) information, can be used in HRRP-based RATR. Actually, the measurement conditions are difficult to be kept exactly same in applications. The amplitude-scale of the received HRRP is a function of radar transmit power, antenna gain, AGC and STC parameter, etc. Therefore, usually only the waveforms of HRRPs without amplitude-scale sensitivity can be used in HRRP target recognition. This is accomplished by amplitude normalization, e.g., normalization.

5. Experimental Results

5.1 Data Description

The experiments presented in this paper are based on the measured data of three types of planes. The parameters of radar and planes are given in Table 1 and the projections of plane trajectories onto ground plane are segmented as displayed in Fig. 2. We take the 5th and the 6th segments of An-26, the 6th and the 7th segments of Cessna and the 2nd and the 5th segments of Yark-42 as training samples, while the remaining data are left for testing. These training data almost cover all of the target-aspect angles. There are 25600 HRRP samples in each segment except the 5th segment of Yark-42 which has 10240 HRRPs. The training data are divided into aspect-frames by the equal interval partition method with each frame containing 1024 HRRP samples.

Thus there are 50, 50 and 35 frames for An-26, Cessna and Yark-42, respectively. In addition, the dimensionality of the measured HRRP is 128.

Table 1. Parameters of radar and planes

Radar Parameters	Frequency	5520 MHz		
	Bandwidth	400 MHz		
	PRF	400 Hz		
	Length (m)	Width (m)	Height	
Yark-42	36.38	34.88	9.83	
An-26	23.80	29.20	9.83	
Cessna	14.40	15.90	9.83	

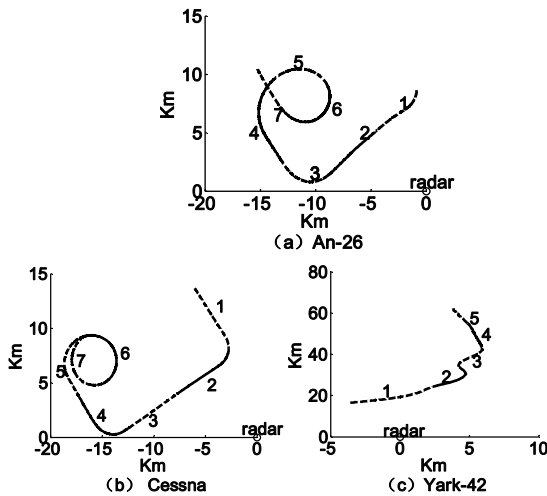


Fig. 2. The projections of three target trajectories onto the ground plane.

5.2 Recognition results

In Table.2 we give the mean and standard deviation values of average correct recognition rate (ACRR), training time, test time for different algorithms. From which we can see that along with complexity of the statistical model increasing, the ACRR increases, but the training and test time also increased.

In Fig.3, we give the ACRR for different recognition algorithms for different SNR level. From which we can see that the ACRR goes down very fast along with the SNR goes lower for the original statistical model. But for the adaptive algorithm, the performance is much robust with the SNR variation.

Table 2 Mean and standard deviation values of ACRR(in percentage), time cost for training and test

	ACRR	training time/min	test time/s
FA	94.1±3.0	4.9	0.4
LFA	95.0±2.6	22.4	2.1
TFA	95.5±2.0	15.2	3.3

6. Conclusion

We give a brief overview for the statistical modelling algorithm for HRRP recognition and some results in this

paper. The performance of these algorithms need to be further evaluated by using more data.

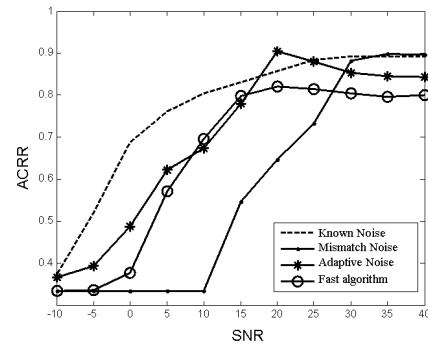


Fig. 3. The recognition performance for different algorithms under noisy case

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